Project 3 Harmonic Analysis

Simon Gustafsson

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1 Introduction

Definition 1 (Weights)**.** *A* weight *is a function $w \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$ with $w \geq 0$. For a measurable set $E \subseteq \mathbb{R}^d$ we write $w(E) := \int_E w(x)\,dx$.*

Definition 2 (Weighted L^p spaces)**.** *Let w be a weight and let $1 \leq p < \infty$. We define*

$$L^p_w(\mathbb{R}^d) := \left\{ f : \mathbb{R}^d \to \mathbb{R} \text{ measurable} : \int_{\mathbb{R}^d} |f(x)|^p\, w(x)\,dx < \infty \right\},$$

with norm $\|f\|_{L^p_w} := \left(\int_{\mathbb{R}^d} |f(x)|^p\, w(x)\,dx\right)^{1/p}$. Moreover, $L^\infty_w(\mathbb{R}^d)$ is the space of measurable functions f with $\|f\|_{L^\infty_w} := \inf\{t \geq 0 : w(\{|f| > t\}) = 0\} < \infty$.

The central object of this report is the Hardy–Littlewood maximal operator M, which assigns to each point the largest average of $|f|$ over cubes containing that point. In the proof of our main result we will also need the analogous operator in which these averages are taken with respect to the measure $w(x)\,dx$ instead of Lebesgue measure.

Definition 3 (Hardy–Littlewood maximal operator)**.** *Let w be a weight with $w(Q) > 0$ for every cube Q. For a measurable function f and $x \in \mathbb{R}^d$,*

$$M_w f(x) := \sup_{Q \ni x} \frac{1}{w(Q)} \int_Q |f(y)|\, w(y)\,dy,$$

where the supremum is taken over all cubes $Q \subset \mathbb{R}^d$ with sides parallel to the axes containing x. If the supremum is taken only over the cubes $Q(x,r)$, $r > 0$, where $Q(x,r)$ denotes the cube centred at x with side length r, we write M^c_w. If $w \equiv 1$ we drop the subscript and write M and M^c.

It is well known that M is of weak type $(1,1)$ and strong type (p,p) on $L^p(\mathbb{R}^d)$ for $1 < p \leq \infty$ [4]. In the light of our $L^p_w(\mathbb{R}^d)$ definition a natural question arises, for fixed $1 < p < \infty$, for which weights w is the sublinear operator M bounded on $L^p_w(\mathbb{R}^d)$? In other words, for which w is there a constant C such that for all $f \in L^p_w(\mathbb{R}^d)$

$$\|Mf\|_{L^p_w} = \left(\int_{\mathbb{R}^d} (Mf)^p\, w\,dx\right)^{1/p} \leq C \left(\int_{\mathbb{R}^d} |f|^p\, w\,dx\right)^{1/p} = C\,\|f\|_{L^p_w}. \tag{1}$$

Remark 1 (The zero weight)**.** *If $w = 0$ almost everywhere, then $\|f\|_{L^p_w} = 0$ for every f, so $L^p_w(\mathbb{R}^d)$ is trivial and M is trivially bounded.*

Remark 2 (The endpoint $p = \infty$)**.** *For $p = \infty$ the weight plays essentially no role. If $w > 0$ almost everywhere, then $w(E) = 0$ if and only if $|E| = 0$, so $L^\infty_w(\mathbb{R}^d) = L^\infty(\mathbb{R}^d)$ with equal norms, and M is bounded by the unweighted result. If w is not almost everywhere zero but vanishes on a set Z with $|Z| > 0$, then $\|\chi_Z\|_{L^\infty_w} = 0$ while $M\chi_Z > 0$ everywhere, so M is not bounded. This is why the question* (1) *is posed only for $1 < p < \infty$.*

Suppose that (1) holds for a weight w. A first observation is that w cannot vanish on a set of positive measure. The indicator function of a bounded part of such a set has norm zero in L^p_w, while its maximal function is positive everywhere, which contradicts (1). More generally, w cannot be much smaller on part of a cube than on the rest of it. A function concentrated where w is small has a small norm in L^p_w, while its maximal function is large on the whole cube, where w is large. Making this precise leads to the following necessary condition.

Proposition 1. *Let $1 < p < \infty$ and let w be a weight that is not almost everywhere zero. If* (1) *holds with constant C, then for every cube Q we have $\int_Q w^{-\frac{1}{p-1}}\,dx < \infty$ and*

$$\left(\frac{1}{|Q|}\int_Q w\,dx\right)\left(\frac{1}{|Q|}\int_Q w^{-\frac{1}{p-1}}\,dx\right)^{p-1} \leq C^p. \tag{2}$$

In particular, $w > 0$ almost everywhere.

Thus, if (1) holds for a weight w, then by Proposition 1, w satisfies (2) for every cube Q, with the same constant C.This gives rise to the next definition by Muckenhoupt [6].

Definition 4 (Muckenhoupt A_p weights)**.** *Let $1 < p < \infty$ and let w be a weight. The A_p* characteristic *of w is*

$$[w]_{A_p} := \sup_Q \left(\frac{1}{|Q|}\int_Q w(x)\,dx\right)\left(\frac{1}{|Q|}\int_Q w(x)^{-\frac{1}{p-1}}\,dx\right)^{p-1}, \tag{3}$$

where the supremum is taken over all cubes $Q \subset \mathbb{R}^d$. We say that $w \in A_p$ if $[w]_{A_p} < \infty$.

Remark 3 (The dual weight)**.** *For a weight w and $1 < p < \infty$ we write $\sigma := w^{-\frac{1}{p-1}}$, with $\sigma = \infty$ where $w = 0$, and $\sigma(E) := \int_E \sigma\,dx$ for measurable sets E. With this notation* (3) *takes the more compact form*

$$[w]_{A_p} = \sup_Q \frac{w(Q)}{|Q|}\left(\frac{\sigma(Q)}{|Q|}\right)^{p-1}.$$

Taking the supremum over all cubes in Proposition 1 gives the following.

Corollary 1. *Let $1 < p < \infty$ and let w be a weight that is not almost everywhere zero. If* (1) *holds with constant C, then $w \in A_p$ and*

$$[w]_{A_p} \leq C^p. \tag{4}$$

Corollary 1 shows that $w \in A_p$ is a necessary condition for (1) to hold. It is natural to ask whether it is also sufficient, that is, whether for every $w \in A_p$ there is a constant C such that (1) holds. Since Corollary 1 moreover shows that any such C must satisfy $C \geq [w]^{1/p}_{A_p}$, we may ask more precisely how large C has to be in terms of $[w]_{A_p}$. The following theorem, due to Buckley [1], answers both questions.

Theorem 1 (Buckley; Theorem C in [5]). *Let $1 < p < \infty$. There is a constant $c_{p,d}$, depending only on p and d, such that for every $w \in A_p$ and every $f \in L^p_w(\mathbb{R}^d)$,*

$$\|Mf\|_{L^p_w} \le c_{p,d}\,[w]_{A_p}^{\frac{1}{p-1}}\,\|f\|_{L^p_w}.$$

The exponent in Theorem 1 cannot be improved.

Proposition 2 (Buckley [1]). *Let $1 < p < \infty$ and $\alpha < \frac{1}{p-1}$. Then there is no constant c such that $\|Mf\|_{L^p_w} \le c\,[w]_{A_p}^{\alpha}\,\|f\|_{L^p_w}$ for all $w \in A_p$ and all $f \in L^p_w(\mathbb{R}^d)$.*

In particular, Theorem 1 shows that $w \in A_p$ is also a sufficient condition, since if $[w]_{A_p} < \infty$, then (1) holds with $C = c_{p,d}[w]_{A_p}^{1/(p-1)}$. Together with Corollary 1, this answers the question posed at the beginning completely and implies the following result.

Theorem 2 (Muckenhoupt). *Let $1 < p < \infty$ and let w be a weight that is not almost everywhere zero. Then M is bounded on $L^p_w(\mathbb{R}^d)$ if and only if $w \in A_p$.*

Theorem 2 is the original result of [6], published in 1972. His proof, as well as Buckley's proof of Theorem 1, relies on the fact that every A_p weight belongs to $A_{p-\varepsilon}$ for some $\varepsilon > 0$, which is a consequence of the reverse Hölder inequality for A_p weights. Buckley's proof moreover uses interpolation. We present neither of these proofs. For the upper bound in Theorem 1 we instead present the elementary proof of Lerner [5], which avoids the $A_{p-\varepsilon}$ property, and as noted above Theorem 2 then needs no separate proof since it follows immidiatly from Corollary 1 and Theorem 1. We complement Lerner's argument with proofs of Propositions 1 and 2.

2 Proofs

2.1 Proof of Proposition 1

Proof. Let Q be a cube. We first show that $w(Q) > 0$. Since w is not almost everywhere zero, the set $F := \{w > 0\}$ has positive measure, so $w(F) > 0$. Every $x \in \mathbb{R}^d$ lies in a cube Q' containing Q, so $M\chi_Q(x) \ge \frac{|Q|}{|Q'|} > 0$ for every x, and hence

$$\|M\chi_Q\|^p_{L^p_w} \ge \int_F (M\chi_Q)^p\, w\, dx > 0.$$

If $w(Q) = 0$, then $\|\chi_Q\|_{L^p_w} = w(Q)^{1/p} = 0$, which contradicts (1). Thus $w(Q) > 0$.

Now let σ be as in Remark 3, let $N > 0$, and let $f := \min\{\sigma, N\}\,\chi_Q$. Where $w > 0$ we have $\sigma^{p-1}w = 1$, so $(\min\{\sigma, N\})^p\, w \le \min\{\sigma, N\}\,\sigma^{p-1}w = \min\{\sigma, N\}$, and where $w = 0$ we have $(\min\{\sigma, N\})^p\, w = 0 \le \min\{\sigma, N\}$. Hence

$$\|f\|^p_{L^p_w} = \int_Q (\min\{\sigma, N\})^p\, w\, dx \le \int_Q \min\{\sigma, N\}\, dx \le N|Q| < \infty,$$

so $f \in L^p_w(\mathbb{R}^d)$. Moreover, Q is a cube containing every $x \in Q$, so $Mf(x) \ge \frac{1}{|Q|}\int_Q \min\{\sigma, N\}\, dy$ for $x \in Q$, and therefore

$$\|Mf\|^p_{L^p_w} \ge \int_Q \left(\frac{1}{|Q|}\int_Q \min\{\sigma, N\}\, dy\right)^p w\, dx = \left(\frac{1}{|Q|}\int_Q \min\{\sigma, N\}\, dy\right)^p w(Q).$$

Combining these two estimates with (1) gives

$$\left(\frac{1}{|Q|}\int_Q \min\{\sigma, N\}\,dy\right)^p w(Q) \le C^p \int_Q \min\{\sigma, N\}\,dy.$$

Since w is locally integrable, $w < \infty$ almost everywhere, so $\min\{\sigma, N\} > 0$ almost everywhere and $\int_Q \min\{\sigma, N\}\,dy > 0$. Dividing by $\int_Q \min\{\sigma, N\}\,dy$ we obtain

$$\frac{w(Q)}{|Q|}\left(\frac{1}{|Q|}\int_Q \min\{\sigma, N\}\,dy\right)^{p-1} \le C^p.$$

As $N \to \infty$, $\min\{\sigma, N\}$ increases to σ, so by monotone convergence $\int_Q \min\{\sigma, N\}\,dy \to \sigma(Q)$, and (2) follows. Since $w(Q) > 0$, this also shows that $\sigma(Q) < \infty$. Finally, $\sigma = \infty$ on $\{w = 0\}$, so $\sigma(Q) < \infty$ implies $|\{w = 0\} \cap Q| = 0$. As this holds for every cube Q, we conclude that $w > 0$ almost everywhere. □

2.2 Preliminaries

Lemma 1. *Let Q be a cube with side length $\ell(Q)$ and let $y \in Q$. Then $Q \subseteq Q(y, 2\ell(Q)) \subseteq 3Q$, where $3Q$ denotes the cube with the same centre as Q and side length $3\ell(Q)$.*

Proof. Every $z \in Q$ satisfies $|z_i - y_i| \le \ell(Q)$ for $i = 1, \dots, d$, since both z and y lie in Q. Hence $Q \subseteq Q(y, 2\ell(Q))$. Let c be the centre of Q. Then $|y_i - c_i| \le \frac{1}{2}\ell(Q)$ for $i = 1, \dots, d$, so every $z \in Q(y, 2\ell(Q))$ satisfies

$$|z_i - c_i| \le |z_i - y_i| + |y_i - c_i| \le \ell(Q) + \tfrac{1}{2}\ell(Q) = \tfrac{3}{2}\ell(Q)$$

for $i = 1, \dots, d$, that is, $z \in 3Q$. □

Lemma 2. *Let $f \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$. Then $Mf(x) \le 2^d M^c f(x)$ for every $x \in \mathbb{R}^d$.*

Proof. Let $x \in \mathbb{R}^d$. For a cube Q containing x, Lemma 1 gives $Q \subseteq Q(x, 2\ell(Q))$, and since $|Q(x, 2\ell(Q))| = 2^d|Q|$ we obtain

$$Mf(x) = \sup_{Q \ni x} \frac{1}{|Q|}\int_Q |f(y)|\,dy \le \sup_{Q \ni x} \frac{2^d}{|Q(x, 2\ell(Q))|}\int_{Q(x,2\ell(Q))} |f(y)|\,dy \le 2^d M^c f(x),$$

where the last inequality holds because each $Q(x, 2\ell(Q))$ is one of the cubes in the definition of $M^c f(x)$. □

Lemma 3. *Let $1 < p < \infty$ and let $w \in A_p$ be a weight that is not almost everywhere zero. Then $w > 0$ almost everywhere, and σ is a weight. Moreover $w(Q) > 0$ and $0 < \sigma(Q) < \infty$ for every cube Q.*

Proof. Suppose that $w = 0$ on a set E with $|E| > 0$. Since w is not almost everywhere zero, there is also a set F with $|F| > 0$ on which $w > 0$. Choose bounded subsets $E' \subseteq E$ and $F' \subseteq F$ of positive measure, and a cube Q containing both. Then $w(Q) \ge w(F') > 0$, while $\sigma = \infty$ on E', so $\sigma(Q) = \infty$. Hence the quantity in (3) is infinite for this cube, which contradicts $w \in A_p$. Thus $w > 0$ almost everywhere, and in particular $w(Q) > 0$ for every cube Q. Since w is locally integrable, $w < \infty$ almost everywhere, so $\sigma > 0$ almost everywhere and $\sigma(Q) > 0$ for every cube Q. Finally, by the definition of $[w]_{A_p}$,

$$\left(\frac{\sigma(Q)}{|Q|}\right)^{p-1} \le [w]_{A_p}\frac{|Q|}{w(Q)} < \infty$$

for every cube Q. Since every compact set is contained in a cube, σ is locally integrable. □

Lemma 4 (Besicovitch covering lemma [3, Lemma A.5])**.** *There is a constant N_d, depending only on d, with the following property. Let $E \subset \mathbb{R}^d$ be a bounded set and, for each $x \in E$, let $r_x > 0$, such that $\sup_{x\in E} r_x < \infty$. Then there are points $x_1, x_2, \ldots \in E$ such that the cubes $Q(x_j, r_{x_j})$ cover E and satisfy $\sum_j \chi_{Q(x_j,r_{x_j})} \le N_d$.*

We use Lemma 4 without proof. A proof can be found in [2, pp. 6–7].

Lemma 5. *Let v be a weight with $v(Q) > 0$ for every cube Q, and let f be a measurable function such that $|f|\,v$ is locally integrable. Then $M_v^c f$ is measurable, and*

$$v\big(\{x \in \mathbb{R}^d : M_v^c f(x) > \lambda\}\big) \le \frac{N_d}{\lambda}\,\|f\|_{L^1_v}$$

for every $\lambda > 0$, where N_d is the constant in Lemma 4.

Proof. Fix $\lambda > 0$, and call a cube $Q(x, r)$ *good* if

$$\int_{Q(x,r)} |f(y)|\,v(y)\,dy > \lambda\, v(Q(x, r)). \tag{5}$$

By the definition of M_v^c, we have $M_v^c f(x) > \lambda$ if and only if $Q(x, r)$ is good for some $r > 0$. For fixed $r > 0$, both sides of (5) are continuous functions of x. Indeed, if $x' \to x$, then $\chi_{Q(x',r)} \to \chi_{Q(x,r)}$ outside the boundary of $Q(x, r)$, which has measure zero, and $\chi_{Q(x',r)} \le \chi_{Q(x,r+2)}$ whenever $|x' - x| \le 1$, so the claim follows by dominated convergence since $|f|\,v$ and v are integrable on $Q(x, r + 2)$. Hence the set U_r of all x for which $Q(x, r)$ is good is open, and so is

$$E_k := \{x : |x| < k\} \cap \bigcup_{0<r\le k} U_r$$

for every positive integer k. We have $E_1 \subseteq E_2 \subseteq \cdots$, every point of E_k has a good cube, and if $Q(x, r)$ is good, then $x \in E_k$ for every integer k with $k > |x|$ and $k \ge r$. Hence $\{M_v^c f > \lambda\} = \bigcup_k E_k$. In particular this set is open, and since $\lambda > 0$ was arbitrary and $M_v^c f \ge 0$, the function $M_v^c f$ is measurable.

For each $x \in E_k$ we choose $r_x \in (0, k]$ such that $Q(x, r_x)$ is good. Since E_k is bounded and $r_x \le k$ for every $x \in E_k$, Lemma 4 gives points $x_1, x_2, \ldots \in E_k$ such that the cubes $Q(x_j, r_{x_j})$ cover E_k and satisfy $\sum_j \chi_{Q(x_j,r_{x_j})} \le N_d$. Using (5) for each $Q(x_j, r_{x_j})$ and monotone convergence to interchange sum and integral, we obtain

$$\begin{aligned} v(E_k) &\le \sum_j v\big(Q(x_j, r_{x_j})\big) \le \frac{1}{\lambda} \sum_j \int_{Q(x_j,r_{x_j})} |f|\,v\,dy \\ &= \frac{1}{\lambda} \int_{\mathbb{R}^d} \Big(\sum_j \chi_{Q(x_j,r_{x_j})}\Big) |f|\,v\,dy \le \frac{N_d}{\lambda}\,\|f\|_{L^1_v}. \end{aligned}$$

Since $E_1 \subseteq E_2 \subseteq \cdots$ and $\bigcup_k E_k = \{M_v^c f > \lambda\}$, continuity of the measure $v(x)\,dx$ from below gives the claim. □

Lemma 6. *Let v be a weight with $v(Q) > 0$ for every cube Q, and let $1 < q < \infty$. Then there is a constant $C_{q,d}$, depending only on q and d, such that*

$$\|M_v^c f\|_{L^q_v} \le C_{q,d}\,\|f\|_{L^q_v}$$

for all $f \in L^q_v(\mathbb{R}^d)$.

Proof. If $f = f_1 + f_2$ with $f_1 \in L^1_v(\mathbb{R}^d)$ and $f_2 \in L^\infty_v(\mathbb{R}^d)$, then $\int_Q |f|\, v\, dy \le \|f_1\|_{L^1_v} + \|f_2\|_{L^\infty_v}\, v(Q) < \infty$ for every cube Q. Hence M^c_v is defined on $L^1_v(\mathbb{R}^d) + L^\infty_v(\mathbb{R}^d)$, and $M^c_v f$ is measurable by Lemma 5. Since $|f+h| \le |f| + |h|$, the operator M^c_v is sublinear.

By Lemma 5, $v(\{M^c_v f > \lambda\}) \le \frac{N_d}{\lambda}\|f\|_{L^1_v}$ for every $f \in L^1_v(\mathbb{R}^d)$ and $\lambda > 0$. Moreover, if $f \in L^\infty_v(\mathbb{R}^d)$, then $|f| \le \|f\|_{L^\infty_v}$ outside a set of v-measure zero, so every average in the definition of $M^c_v f$ is at most $\|f\|_{L^\infty_v}$, and $\|M^c_v f\|_{L^\infty_v} \le \|f\|_{L^\infty_v}$. The Marcinkiewicz interpolation theorem [3, Theorem A.6], applied on the measure space $(\mathbb{R}^d, v(x)\,dx)$ with exponents 1 and ∞, now gives the claim with a constant depending only on q, N_d and 1, and hence only on q and d. □

2.3 Proof of Theorem 1

We prove Theorem 1 following the argument of Lerner [5].

Proof of Theorem 1. Let $1 < p < \infty$, let $w \in A_p$ and let $f \in L^p_w(\mathbb{R}^d)$. If $w = 0$ almost everywhere, both sides of the inequality vanish, so we may assume that w is not almost everywhere zero. By Lemma 3, $w > 0$ almost everywhere and σ is a weight with $w(Q) > 0$ and $0 < \sigma(Q) < \infty$ for every cube Q. In particular, the operators M^c_w and M^c_σ are defined.

We first prove a pointwise estimate for Mf. Fix $x \in \mathbb{R}^d$ and $r > 0$, let $Q := Q(x,r)$, and let $3Q$ be as in Lemma 1. Multiplying and dividing by $w(Q)\,\sigma(3Q)^{p-1}$, we obtain

$$\begin{aligned}\left(\frac{1}{|Q|}\int_Q |f(y)|\,dy\right)^{p-1} &= \frac{1}{|Q|^{p-1}}\left(\int_Q |f(y)|\,dy\right)^{p-1}\\ &= \frac{w(Q)\,\sigma(3Q)^{p-1}}{|Q|^p}\cdot\frac{|Q|}{w(Q)}\cdot\frac{1}{\sigma(3Q)^{p-1}}\left(\int_Q |f(y)|\,dy\right)^{p-1}\\ &= \frac{w(Q)\,\sigma(3Q)^{p-1}}{|Q|^p}\cdot\frac{|Q|}{w(Q)}\left(\frac{1}{\sigma(3Q)}\int_Q |f(y)|\,dy\right)^{p-1}.\end{aligned} \tag{6}$$

Since $w(Q) \le w(3Q)$ and $|3Q| = 3^d|Q|$, the first factor on the right-hand side satisfies

$$\frac{w(Q)\,\sigma(3Q)^{p-1}}{|Q|^p} \le 3^{dp}\left(\frac{1}{|3Q|}\int_{3Q} w(y)\,dy\right)\left(\frac{1}{|3Q|}\int_{3Q}\sigma(y)\,dy\right)^{p-1} \le 3^{dp}\,[w]_{A_p}.$$

For the second factor, let $y \in Q$. By Lemma 1, $Q \subseteq Q(y, 2\ell(Q)) \subseteq 3Q$, so

$$\begin{aligned}\frac{1}{\sigma(3Q)}\int_Q |f(z)|\,dz &= \frac{1}{\sigma(3Q)}\int_Q |f(z)|\,\sigma(z)^{-1}\,\sigma(z)\,dz\\ &\le \frac{1}{\sigma(Q(y,2\ell(Q)))}\int_{Q(y,2\ell(Q))} |f(z)|\,\sigma(z)^{-1}\,\sigma(z)\,dz \le M^c_\sigma\big(|f|\sigma^{-1}\big)(y).\end{aligned}$$

This holds for every $y \in Q$, and therefore

$$\begin{aligned}\frac{|Q|}{w(Q)}\left(\frac{1}{\sigma(3Q)}\int_Q |f(z)|\,dz\right)^{p-1} &= \frac{1}{w(Q)}\int_Q\left(\frac{1}{\sigma(3Q)}\int_Q |f(z)|\,dz\right)^{p-1} dy\\ &\le \frac{1}{w(Q)}\int_Q \Big(M^c_\sigma\big(|f|\sigma^{-1}\big)(y)\Big)^{p-1} w(y)^{-1}\,w(y)\,dy\\ &\le M^c_w\Big(\big(M^c_\sigma(|f|\sigma^{-1})\big)^{p-1} w^{-1}\Big)(x),\end{aligned}$$

where the last inequality holds because $Q = Q(x,r)$. Inserting both bounds into (6), taking the supremum over $r > 0$ and applying Lemma 2, we arrive at

$$Mf(x) \le 2^d\, 3^{\frac{dp}{p-1}}\, [w]_{A_p}^{\frac{1}{p-1}} \Big(M_w^c\Big(\big(M_\sigma^c(|f|\sigma^{-1})\big)^{p-1} w^{-1}\Big)(x)\Big)^{\frac{1}{p-1}}. \tag{7}$$

We now take norms. Since $1 - \frac{p}{p-1} = -\frac{1}{p-1}$, we have $w^{1-\frac{p}{p-1}} = \sigma$, and similarly $\sigma^{1-p} = w$. The second identity gives

$$\big\| |f|\sigma^{-1}\big\|_{L^p_\sigma}^p = \int_{\mathbb{R}^d} |f(x)|^p\, \sigma(x)^{-p}\, \sigma(x)\, dx = \int_{\mathbb{R}^d} |f(x)|^p\, w(x)\, dx = \|f\|_{L^p_w}^p,$$

so $|f|\sigma^{-1} \in L^p_\sigma(\mathbb{R}^d)$. Using the first identity and Lemma 6 with $v = \sigma$ and $q = p$, we obtain

$$\begin{aligned}\int_{\mathbb{R}^d} \Big(\big(M_\sigma^c(|f|\sigma^{-1})(x)\big)^{p-1} w(x)^{-1}\Big)^{\frac{p}{p-1}} w(x)\, dx &= \int_{\mathbb{R}^d} \big(M_\sigma^c(|f|\sigma^{-1})(x)\big)^p\, \sigma(x)\, dx \\ &\le C_{p,d}^p\, \|f\|_{L^p_w}^p,\end{aligned}$$

so the function $\big(M_\sigma^c(|f|\sigma^{-1})\big)^{p-1} w^{-1}$ lies in $L^{p/(p-1)}_w(\mathbb{R}^d)$. Finally, we raise (7) to the power p, integrate against w, and apply Lemma 6 with $v = w$ and $q = \frac{p}{p-1}$. Together with the previous estimate this gives

$$\|Mf\|_{L^p_w}^p \le \Big(2^d\, 3^{\frac{dp}{p-1}}\Big)^p\, [w]_{A_p}^{\frac{p}{p-1}}\, C_{\frac{p}{p-1},d}^{\frac{p}{p-1}}\, C_{p,d}^p\, \|f\|_{L^p_w}^p.$$

Taking p-th roots, we conclude that

$$\|Mf\|_{L^p_w} \le c_{p,d}\, [w]_{A_p}^{\frac{1}{p-1}}\, \|f\|_{L^p_w}, \qquad c_{p,d} := 2^d\, 3^{\frac{dp}{p-1}}\, C_{\frac{p}{p-1},d}^{\frac{1}{p-1}}\, C_{p,d},$$

and $c_{p,d}$ depends only on p and d. □

2.4 Sharpness of the exponent

Proof of Proposition 2. Let $1 < p < \infty$ and $\alpha < \frac{1}{p-1}$, and suppose, for contradiction, that there is a constant c such that

$$\|Mf\|_{L^p_w} \le c\, [w]_{A_p}^{\alpha}\, \|f\|_{L^p_w} \tag{8}$$

for all $w \in A_p$ and all $f \in L^p_w(\mathbb{R}^d)$. For every weight w that is not almost everywhere zero and every cube Q, Hölder's inequality with exponents p and $\frac{p}{p-1}$ gives

$$1 = \frac{1}{|Q|}\int_Q w^{\frac{1}{p}}\, w^{-\frac{1}{p}}\, dy \le \left(\frac{1}{|Q|}\int_Q w\, dy\right)^{\frac{1}{p}} \left(\frac{1}{|Q|}\int_Q w^{-\frac{1}{p-1}}\, dy\right)^{\frac{p-1}{p}},$$

so $[w]_{A_p} \ge 1$. Hence (8) remains true if α is replaced by $\max\{\alpha, 0\}$, and we may assume that $0 \le \alpha < \frac{1}{p-1}$.

For $0 < \delta < 1$ we consider the weight $w_\delta(x) := |x|^{(d-\delta)(p-1)}$, for which $w_\delta^{-\frac{1}{p-1}}(x) = |x|^{\delta-d}$. By polar coordinates,

$$\int_{\{|y|<r\}} |y|^{\delta-d}\, dy = \omega_d\, \frac{r^\delta}{\delta} \qquad \text{for every } r > 0, \tag{9}$$

where ω_d denotes the surface measure of the unit sphere in $\mathbb{R}^d$.

We first show that $w_\delta \in A_p$ with

$$[w_\delta]_{A_p} \le C\,\delta^{1-p}, \tag{10}$$

where C depends only on p and d. The function w_δ is continuous and positive outside the origin, so it is a weight that is not almost everywhere zero. Write $a := (d-\delta)(p-1) \ge 0$, and let $Q := Q(c,\ell)$ be a cube. Every $y \in Q$ satisfies $|y-c| \le \frac{\sqrt{d}}{2}\,\ell$.

Suppose first that $|c| \le \sqrt{d}\,\ell$. Then every $y \in Q$ satisfies $|y| \le R\ell$ with $R := \frac{3\sqrt{d}}{2}$. Since $a \ge 0$, this gives $w_\delta \le (R\ell)^a$ on Q, and by (9),

$$\frac{1}{|Q|}\int_Q |y|^{\delta-d}\,dy \le \frac{1}{\ell^d}\int_{\{|y|<R\ell\}} |y|^{\delta-d}\,dy = \frac{\omega_d\,R^\delta\,\ell^{\delta-d}}{\delta}.$$

Since $a + (\delta-d)(p-1) = 0$, we obtain

$$\left(\frac{1}{|Q|}\int_Q w_\delta\,dy\right)\left(\frac{1}{|Q|}\int_Q |y|^{\delta-d}\,dy\right)^{p-1} \le (R\ell)^a\left(\frac{\omega_d\,R^\delta\,\ell^{\delta-d}}{\delta}\right)^{p-1} = R^{d(p-1)}\,\omega_d^{p-1}\,\delta^{1-p}.$$

Suppose instead that $|c| > \sqrt{d}\,\ell$. Then $|y-c| < \frac{1}{2}|c|$, so $\frac{1}{2}|c| \le |y| \le \frac{3}{2}|c|$ for every $y \in Q$. Hence $w_\delta \le \left(\frac{3}{2}|c|\right)^a$ and $|y|^{\delta-d} \le \left(\frac{1}{2}|c|\right)^{\delta-d}$ on Q, and

$$\left(\frac{1}{|Q|}\int_Q w_\delta\,dy\right)\left(\frac{1}{|Q|}\int_Q |y|^{\delta-d}\,dy\right)^{p-1} \le \left(\tfrac{3}{2}\right)^a 2^{(d-\delta)(p-1)} = 3^a \le 3^{d(p-1)} \le 3^{d(p-1)}\,\delta^{1-p}.$$

Taking the supremum over all cubes proves (10) with $C := \max\{R^{d(p-1)}\omega_d^{p-1},\, 3^{d(p-1)}\}$.

Next we construct a function for which M is large in $L^p_{w_\delta}$. Let $f_\delta(x) := |x|^{\delta-d}\chi_{Q(0,2)}(x)$. Then $f_\delta \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$, and since $Q(0,2) \subseteq \{|y| < 2\sqrt{d}\}$, (9) gives

$$\|f_\delta\|^p_{L^p_{w_\delta}} = \int_{Q(0,2)} |y|^{(\delta-d)p}\,|y|^{(d-\delta)(p-1)}\,dy = \int_{Q(0,2)} |y|^{\delta-d}\,dy \le \omega_d\,\frac{(2\sqrt{d})^\delta}{\delta} < \infty.$$

Moreover $\|f_\delta\|_{L^p_{w_\delta}} > 0$, so $f_\delta \in L^p_{w_\delta}(\mathbb{R}^d)$ is not the zero element. Let $x \in Q(0,2)$ with $x \ne 0$, and put $r := \max_{1\le i\le d}|x_i|$, so that $0 < r \le |x|$. The cube $Q(0,2r)$ contains x, is contained in $Q(0,2)$, and contains the ball $\{|y| < r\}$. By the definition of M and (9),

$$Mf_\delta(x) \ge \frac{1}{(2r)^d}\int_{Q(0,2r)} |y|^{\delta-d}\,dy \ge \frac{1}{(2r)^d}\int_{\{|y|<r\}} |y|^{\delta-d}\,dy = \frac{\omega_d}{2^d\delta}\,r^{\delta-d} \ge \frac{\omega_d}{2^d\delta}\,|x|^{\delta-d},$$

where the last inequality holds because $\delta - d < 0$ and $r \le |x|$. Thus $Mf_\delta \ge \frac{\omega_d}{2^d\delta} f_\delta$ almost everywhere on $Q(0,2)$, and this also holds outside $Q(0,2)$, where $f_\delta = 0$. Consequently

$$\|Mf_\delta\|_{L^p_{w_\delta}} \ge \frac{\omega_d}{2^d\delta}\,\|f_\delta\|_{L^p_{w_\delta}}. \tag{11}$$

Finally we combine these estimates. Applying (8) to w_δ and f_δ, and using (11), (10) and $\alpha \ge 0$, we obtain

$$\frac{\omega_d}{2^d\delta}\,\|f_\delta\|_{L^p_{w_\delta}} \le \|Mf_\delta\|_{L^p_{w_\delta}} \le c\,[w_\delta]^\alpha_{A_p}\,\|f_\delta\|_{L^p_{w_\delta}} \le c\,C^\alpha\,\delta^{\alpha(1-p)}\,\|f_\delta\|_{L^p_{w_\delta}}.$$

Dividing by $\|f_\delta\|_{L^p_{w_\delta}} \in (0,\infty)$ and multiplying by δ gives

$$\frac{\omega_d}{2^d} \le c\,C^\alpha\,\delta^{1-\alpha(p-1)} \qquad \text{for every } 0 < \delta < 1.$$

Since $\alpha < \frac{1}{p-1}$, the exponent $1-\alpha(p-1)$ is positive, so the right-hand side tends to 0 as $\delta \to 0$, while the left-hand side is a positive constant independent of δ. This is a contradiction, and the proof is complete. $\square$

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